

Complex radar frame X

Rank- k decomposition
residual R_k

background-tail control

Residual statistic
 $S_{\text{res}} = \sum_c |R_{k,c}|^2$

two score maps; evaluation readout declared separately

Raw statistic
 $S_{\text{raw}} = \sum_c |x_{c,l}|^2$

target-preserving route

Compact gate
 $S_{\text{gate}} = g_\phi(\ell_{\text{raw}}, \ell_{\text{res}})$
 $\ell_\bullet = \log(1 + S_\bullet)$

PRIMARY: target-blind alarm budget
within-frame ranks $Q_{\text{res}}, Q_{\text{gate}}$; $Q_{\text{TP}} = \max(Q_{\text{res}}, Q_{\text{gate}})$
select the common top- κ_α cells

alternative readout of the same score maps

DIAGNOSTIC ONLY: mask-assisted branch union
 $S_{\text{res}} > \tau_{\text{res}, \alpha}$ OR $S_{\text{gate}} > \tau_{\text{gate}, 0}$